

Replacing a Fast Transport Model With a MuJoCo Contact Model Turned a Correct PLC Ladder From [PASS] to [FAIL] in 4 of 12 Simulated Sort-Cell Batches, All With Five Defective Parts; Re-Deriving Three Ladder Constants in MuJoCo Removed None of the Four Reversals, and Detection of Known Ladder Defects Did Not Change: A Pre-Registered Comparison in Which All Eight Predictions Held

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Released 2026-10-08. Numbers are from the second official round tr05-official-2 (2026-10-08 14:30:33–17:25:56 KST), the only valid one. The prediction freeze was committed to the public repository before any calibration or official run (waceinc/tech-reports commit 3a12187, 2026-10-08 13:33:43 KST); amendment 1 (1ac8c13), the fixed version v1.0 (1ee07b3) and correction 1 (cd0d54a) followed, all before the second round. Correction 2 (one line of the independent recompute script) was committed after the official results (2278552, 2026-10-08 18:04:16 KST). Not peer-reviewed; reviews were done by AI agents, not by human reviewers.

Abstract

A PLC ladder is often checked against a plant model before it meets equipment, and the verdict depends on that model. We asked how often the verdict of the same ladder changes when only the way parts are moved is changed, in two simulated conveyor cells: a shuttle cell (cell A) and a sort cell with a pusher and an inclined chute (cell B(3i)). Condition F moves parts with fast kinematic rules; condition M0 computes the same parts as rigid boxes in MuJoCo, with the same actuators, sensors and ladder; condition M1 is M0 with three sort-cell ladder constants re-derived from MuJoCo by a rule fixed before calibration. Before any official run we registered eight predictions and three falsification conditions. The official round had 607 runs, all valid; the 115 pairs that were run twice (9 fast-model gate pairs and 106 MuJoCo pairs) were identical. In the sort cell, the correct ladder passed under F in all 12 batches of the `normal` scenario and failed under M0 in 4, all four in the batches with five defective parts (first lock: chute full in 3, chute confirm window in 1). The same 4 failed under M1, now all with chute full, so the re-derived constants did not remove any of them. In the 13 `curtain` and `recovery` checks of the correct ladder, F and M0 agreed. Detection of known defects by the expected reason did not change between F and M0 (21 of 28 pairs in both, no discordant pair); the 7 misses were the same 7 batches in all conditions and, by the unreleased tick records, come from the scenario's trigger, not from the plant model. **All eight predictions held and no falsification condition was triggered.** Two of them say less than the count suggests: P3 held largely by construction, because the re-derivation does not change the chute capacity on which a chute-full lock depends, and P8 counted as holding two settings that made every batch fail, including the eight that agree at baseline. The result describes two simulator models of the same cells on one machine and one build; it does not say which model is closer to a physical conveyor, and no physical PLC or conveyor was compared. The reversals are sensitive to assumed physical values: lowering friction by 30 % removed all four, and raising it by 30 % made all 12 batches fail. The sort cell here is the 3i design and ladder, not the 3k cell of TR-2026-03 and TR-2026-04 (TR-2026-02 ran cell A only), so its results cannot be read against those reports. The application source is not public; the release bundle holds the pre-registration with its amendment and corrections, the public result files and the scripts that recompute every verdict.

1. Question

A ladder that drives a simulated plant is judged by what the plant model makes happen: which sensors switch when, whether parts reach the right exit, whether the ladder locks out. The model that moves the parts is therefore part of the test. We asked, before any official run, with the same cell definition, the same ladder and the same stimuli:

1. **Defect detection.** Of the ladders with a known injected defect (negative ladders), how many does each model detect for the expected reason, and how many pairs disagree?
2. **Correct-ladder reversals.** How often does the verdict of the correct ladder ([PASS] or [FAIL]) change with the model, in which direction, and what is the first event that differs?
3. **After re-derivation.** Does a reversal remain when the correct-ladder constants that were originally fitted to the fast model's history are re-derived from MuJoCo (condition M1)?

"The MuJoCo model" is a bundle, not only the contact solver: MuJoCo rigid-body contact, the transport rules that carry fast-model rules over to it (Section 3.2), the assumed physical values (Appendix B), and the absence of some fast-model rules (Appendix C). The confirmatory claim is that *this bundle* changes verdicts; splitting a reversal into "physics" and "port, rule or bookkeeping differences" is done only by the pre-registered classification and mapping (Section 5.1), not as a confirmed claim. **Which model is right is not decided.** There is no physical cell, so the report does not claim that MuJoCo is more realistic.

2. Why this is a simulation-only question

The same ladder, stimuli and part placement can be run side by side with only the transport model changed; the actuators (pusher cylinder trajectory, belt motor, sensor logic) use the same code in both models, and runs within one build were bit-reproducible in this round (P6). The comparison between models takes place inside one simulator on one machine. Absolute verdicts depend on the cell definitions and on assumed physical values and do not describe any site. The report says nothing about physical PLCs, conveyors, sorters or boxes.

3. Experimental design

3.1 Code under test and the two plant models

The code under test is pre-release development code of a WACE Inc. simulation application (development branch at commit `ac78d02fbc84b35f041f4fcc82047a0c80c67930`, not merged into the main line, not shipped). Nothing in this report describes a released product.

Item	Fast model (F)	MuJoCo model (M)
Selection	default	command-line flag only; the runner checks the mode line the application prints at load and the presence of the physics report
Cell A (shuttle, 50 ms plant tick)	belt core moves the part's x directly	static deck box whose surface velocity is set every 1 ms from the core motor's belt displacement, side guides, floor; one dynamic part; 50 fixed steps of 1 ms per tick
Cell B(3i) (sort, 10 ms plant tick)	kinematic rules for belt transport, centering clamp, pusher transfer, arc, chute queue cap, chute brake, output accumulation cap	five deck boxes driven by one conveyor motor (40 mm gaps), centering walls, arc of 10 chord facets, 20° chute, chute side and end walls, output end cushion, floor; 20 dynamic boxes; pusher plate a free dynamic box whose pose and velocity are written from the core cylinder trajectory at every 1 ms boundary (tracking error at most 0.025 mm); 10 steps of 1 ms per tick
Actuators, sensors	core	same core (motors, cylinder timing, reed switches, faults, curtain, panel, photo-sensor logic). Sensors see an axis-aligned box of fixed size at the MuJoCo part centre (rotation not used, except in one sensitivity setting)

Item	Fast model (F)	MuJoCo model (M)
Atomic step	copy plant, advance, commit	same; the copy is a deep clone of the MuJoCo model and data
Numerical problems	—	automatic reset off; any MuJoCo warning in a 1 ms step throws and the tick is not committed (counted as [PLANT_ERROR])
Engine settings	—	MuJoCo 3.12.0, fixed step 0.001 s, implicitfast integrator, elliptic cone, 80 solver iterations, condim 6, solref 0.01, solimp (0.95, 0.99, 0.001) fixed in code, 1 ms edge resolution

3.2 The MuJoCo part-transport rules and model (disclosure)

This section and Appendices B and C describe the transport rules and model in enough detail to serve as their disclosure (see Licence).

- **Mixed plant.** Actuators stay kinematic and come from the same core as the fast model; only part transport and contact are computed in MuJoCo. The pusher plate is a free dynamic box whose position and velocity are overwritten at every 1 ms boundary from the core cylinder trajectory (q_{pos} = core face position, q_{vel} = next face minus this face over 1 ms), so the contact solver sees the plate's velocity; a kinematic (mocap) plate alone let the plate sink centimetres into a part during development. Deck surface velocity is set every 1 ms from the core motor's belt samples.
- **Transport rule v1 (cell A).** The part-drop rule is the fast core's rule: the first 1 ms sample with $|x| > L/2 + 1e-12$ raises PART_DROP (a secondary rule records a side drop when the centre falls below the deck). A part whose route is no longer 0 is frozen (inactive, no collision). Floor top at $y = 0$. Automatic reset off; warnings throw. One part. Tilt and yaw of parts over the deck are reported, with a flag above 2° .
- **Transport rule v4 (cell B(3i)).** Same principles for 20 parts, plus: (a) **chute x force** — every 1 ms each chute part that is not held gets an x-only force $F = m \cdot g \cdot (\mu_x - \mu_{iso} \cdot \cos \theta \cdot |v_x| / |v|)$, clamped to $[0, m \cdot |v_x| / 1 \text{ ms}]$, opposite to v_x , where μ_x is the fast model's chute x friction (0.85), μ_{iso} and θ are the isotropic friction and the local slope of the arc or chute, so that the total x deceleration follows the fast rule while MuJoCo's isotropic friction supplies the rest (the force acts at the centre of mass, so the yaw damping torque of real anisotropic friction is absent); (b) **CHUTE-HOLD** — while the chute-jam fault is on, every chute part is held at its pose; (c) **chute capture** (route 1) by the same criterion as the fast model, at the tick end; (d) **output arrival** (route 2) when the contact stop position is within 2 mm of the fast model's output cap — used only for route counting and the next part's cap, never as a PLC sensor input; (e) **CENTER-SNAP** evaluated on the deepest footprint corner (yaw included) with a 2 mm contact tolerance (measured maximum penetration 1.66 mm); (f) a collected part stays a chute obstacle until it has moved clear, then becomes inactive; (g) an initial contact of a part with anything but a deck throws at load. v4 adds to v3 two sensitivity switches (arc friction scale and chute x force on/off) and the rotated-sensor-footprint switch; their defaults leave the v3 behaviour unchanged.
- **Two hashes.** `values_sha256` covers the rule string, the named physics values, the full compiled scene (MJCF), the compiled options and the engine version, and in cell B every plant parameter (so it differs per seed, belt factor and initial lateral position). `setup_sha256` covers only the rule string, the values, the compiled options and the engine version, so it is one registered value per cell and physics setting. The runner checks both against the pre-registered values in every M run. The scene MJCF is produced by the unreleased transport code; its hash is released, its text is not, so a reader can compare the hashes in the result files with the seed table but cannot recompute them.
- **Reports that no verdict reads.** Per-part pose, velocity, support surface and gap, stacked and on-floor flags every tick; the first plate-to-box contact sample of each pushed box (M from the contact list; F, behind a flag that does not change any other byte of the F response).

Appendix B lists the 15 (cell A) and 32 (cell B(3i)) named physics values with their defaults and sources; Appendix C lists every difference between the two models, row by row.

3.3 Cells and ladders, and why cell B(3i) is not the cell B of TR-2026-03 and TR-2026-04

- **Cell A** (shuttle, one part, PLC scan 10 ms, 5 scans per plant tick): correct ladder (27 rungs, SHA-256 5058bd0b...) and two negative ladders, late-reverse (reversal 150 scans late) and swapped-sensors (left and right sensors swapped), both expected to cause PART_DROP. The correct ladder is the file used in TR-2026-02 to TR-2026-04; the two negative ladders are the files released with TR-2026-02.
- **Cell B(3i)** (sort, 20 parts, defect rate 0.2, PLC scan 10 ms, one scan per tick): the earlier pusher-assembly design (device definition 1.5.0) with its **3i ladder** (788 rungs, eb44ce59...) and three negative ladders, push-timing-off (firing +50 scans; expected a push-timing event), ignore-curtain (expected the bridge judgment PLC_CURTAIN_IGNORED) and no-sort-check (no defect-route check; no expected reason exists anywhere in the 3i bridge, so it counts only in the secondary metric).
- **Not comparable with TR-2026-03 and TR-2026-04.** What those reports call cell B is the later 3k cell (device definition 1.6.0 with a position rule on an accumulating high-speed counter, correct ladder of 569 rungs, 05999e23...), run on macOS; TR-2026-02 dropped cell B before its runs and reports cell A only (the pre-registration's §11 names TR-2026-02 among these reports; it has no cell B results). Cell B(3i) differs in cell design, correct ladder, negative-ladder set and platform (Linux). The MuJoCo mode refuses the 3k design. **The cell B(3i) results of this report must not be pooled with, or read side by side with, the cell B results of TR-2026-03 and TR-2026-04, and they neither correct nor support those reports' conclusions.** Same seed numbers do not mean the same batches. Only the cell A ladder files are shared; even there the platform differs, so cell A results are compared with the earlier reports at the level of verdicts only.

3.4 Conditions and the re-derivation rule (M1)

Condition	Plant	Ladder constants	Reference logic
F	fast	original	original
M0	MuJoCo, default values	original	original
M1	as M0	sort-cell correct ladder and the three negative ladders regenerated with re-derived constants	original logic with the new constants

The 3i constants record their own provenance; three are recorded as fitted to the fast model's history (chute confirm window 165 to 235 scans, chute full ≥ 45 scans, and the subtraction 7 in the firing delay $\lfloor \text{PulseSpan} \times 232 / (i \times 50) \rfloor - 7$). The end-to-end fitting procedure is not documented anywhere, so the pre-registration defines its own rule: on 6 calibration batches, record the F closed-loop run of the correct ladder; replay its requests open-loop on M0, measure the same statistic on both with the original ladder engine, and set the new constant by a fixed formula. The rule ran once, after the freeze (Section 6):

Constant	Statistic (6 calibration batches, 24 defective parts)	New value
Chute confirm window 165–235	age at filtered chute rise: F [195, 196], M [197, 234]; margins kept 30 and 39; median shift +38 scans	167–273
Chute full 45	longest passing sensor-on interval: F 11, M 12 scans	$\lceil 45 \times 12/11 \rceil = 50$
Firing-delay subtraction 7	first plate contact position: median F -0.20420 m, M -0.20902 m, $\Delta x -0.004818$ m	$7 - \lceil 0.004818 / 0.012 \rceil = 6$
Output window 198–269, arrival-on limit 29, output full 100	F and M 233–234; 19; 19 — not near the limits	unchanged (not re-derived by rule)

No calibration seed was skipped. The four M1 ladders differ from the originals only in constant literals (36, 36, 36 and 33 lines). M1 does not change the chute's capacity (sensor position and queue length), which matters for P3 (Section 4). Cell A constants have no recorded source and were not re-derived, so cell A M1 equals M0 and was not run.

3.5 Inputs

- **Cell A:** no seed; scenarios `normal` (1,000 ticks) and `cycle-stop` (1,200 ticks, stop button at tick 600). 3 ladders $\times$ 2 scenarios.
- **Cell B(3i):** official seeds from SHA-256("TR-2026-05/official/" + k), first 4 bytes, k = 1, 2, ..., skipping seeds 0 to 3; defect counts read by loading only (no tick run); the first 4 seeds with 3, 4 and 5 defective parts each — **12 batches**, drawn after the freeze from the committed rule (seed table `d02d62f3...`; the official set needed k = 1 to 35, and 52 values of k were examined across the official, calibration and development sets). Batches with 6 or more defective parts were excluded because the fast model's chute queue can overflow there too. Lateral start position 0.
- **Pairs:** correct ladder $\times$ `normal` (5,000 ticks) $\times$ 12, $\times$ `curtain` (1,000 ticks; operator enters from the tick after the first pusher command) $\times$ 12, $\times$ `recovery` (2,800 ticks) $\times$ 1 (first 5-defect batch); `push-timing-off` $\times$ `normal` (1,800) $\times$ 12; `ignore-curtain` $\times$ `curtain` (500) $\times$ 12; `no-sort-check` $\times$ `flow` (800) $\times$ 1. Main runs: F 56, M0 56, M1 50 = 162.

3.6 Verdicts and measures

- **Process verdict [PASS]** = complete $\wedge$ no physical event (production events excluded) $\wedge$ no bridge judgment. Cell A complete = visits right, left $\times$ 3, Rounds == 3, all outputs off at the end. Cell B complete = every defective part at route 1 or 3, every good part at route 2, Arrivals == 20.
- **Correct curtain:** safety verdict (curtain detected; belt running the tick before; belt and pusher commands off on every curtain tick; stop within 0.265 s; no physical event). **Correct recovery:** the bridge's recovery check.
- **Ledger-free verdict** (reported only): as the process verdict, but a good part at route 0 counts as delivered if it is past the output sensor ($x > 2.2$ m) and not on the floor.
- **Reversal classes** (pre-registered): (a) ladder lock (FaultCode), (b) physical event or bridge judgment, (c) completion items only, (d) other. Reversals for P1, P2, P3, P8, F1, F3 count (a)+(b)+(d). The first differing event is mapped in advance to a row of the difference table (Appendix C).
- **Expected-reason detection** (main metric, 28 pairs: cell A 4, `push-timing-off` 12, `ignore-curtain` 12) = negative ladder [FAIL] $\wedge$ expected reason present. **Verdict detection** (secondary, 29 pairs incl. no `-sort-check`), split into the 13 pairs that fail by run length alone and the other 16.
- **Reference match:** the correct ladder's outputs must equal a reference logic scan by scan up to the first lock (one documented one-scan shape excepted); any other difference invalidates the round.

3.7 Runs, invalidation and positive controls

One Ubuntu machine (Linux aarch64, NVIDIA GB10, Ubuntu 24.04.5, Clang 21.1.8, Release, `-ffp-contract=off`), one executable (`d58c3928...`), `libmujoco ddc46a81...`, Node v24.21.0; runs sequential. The round runner refuses to start if the fixed-input file (`fixed.json`) or any file it lists differs from its SHA-256. Run categories: [VALID], [PLANT_ERROR] (MuJoCo warning; not a verdict), [INVALID]. Every M pair runs twice and the full trace hash is compared; a differing pair would go to a 10-run majority rule. The round is invalid if the F determinism gate (9 pairs, two runs each) differs, if a reference-match difference occurs before the first lock, if more than 10 % of an M condition's runs are [PLANT_ERROR], if a planned run is missing, or if a mode check fails. Further runs per round: 14 open-loop replays (F requests sent to M0 without a ladder), 156 P8 sensitivity runs, $10 \times (f + 2)$ model-assumption runs (f = number of M0 reversals in correct `normal`), and 100 runs of the 25 cell-A interlock-removal variants of TR-2026-03 (E4).

Before the official round, positive control 1 (the fast model reproduces the macOS 3i reference results item by item: 30 of 31 checks, the remaining one being the `no-sort-check` "detection" item that no 3i ladder can pass; the gate file's overall status is [FAIL] for that item, which the macOS reference fails too; the pre-registered criterion is "no-sort-check [FAIL]", which was met) and positive control 2 (3 development seeds through runner, tables, independent recompute and public conversion) were run.

The first positive control 2 was invalid (Section 6). The regression tests counted for positive control 1 were an existing record (11 of 11) of the same executable, not a fresh run.

3.8 What was seen before registration

Before the freeze, development runs used seeds 1 to 3 only. They had shown: the cell A correct ladder [PASS] under F and M0 in both `normal` and `cycle-stop`; the cell B correct ladder with seed 1 (5 defective parts) [PASS] under F and [FAIL] under M0, first lock chute full at tick 1710, with boxes yawing on the chute and the fifth defective box stopping inside the chute sensor band; open-loop sensor edges within 10 ms on the belt and the chute sensor later under M (mean +164 ms). P1 and P7 are reproduction predictions from these observations. **Because cell A has no seed, its official input equals the development input; its correct-ladder results are therefore exploratory (E6) and carry no prediction.** The reference-match invalidation range, the exclusion of `no-sort-check` from the main metric, the defect-count bins and the sensitivity items were set after seeing these runs.

4. Pre-registered predictions and verdicts

Rates are integer comparisons. Missing pairs would be filled both ways; no pair was undecided. The table gives the values and verdicts written by the fixed table script (`tables.py`) and reproduced by the independent recompute; the script writes [맞음] (held) for predictions and [틀림] with "falsification condition not met" for falsifiers.

No.	Prediction	Line	Result	Verdict
P1	Of the 4 five-defect batches, at least 3 with correct normal F [PASS] $\wedge$ M0 [FAIL] $\wedge$ M0 first lock FaultCode 3 (chute full) $\wedge$ class (a)	wrong if 2 or fewer	3 of 4	Held
P2	Of the 8 three- and four-defect batches, at most 2 reversals $F \neq M0$	wrong if 3 or more	0 of 8	Held
P3	Of the $n0$ batches meeting P1, x also F [PASS] $\wedge$ M1 [FAIL]; $2x > n0$ (structural: M1 keeps the chute capacity)	wrong if $2x \leq n0$	$x = 3, n0 = 3$ (all three with FaultCode 3)	Held
P4	Pairs of the main metric (28) where F and M0 disagree on expected-reason detection: at most 1	wrong if 2 or more	0 of 28	Held
P5	The 4 cell A negative pairs detected by PART_DROP under F and M0	wrong if any missed	4 of 4	Held
P6	F gate 9 pairs identical, and every M0 and M1 pair identical in its two runs	wrong if any differs	9 of 9; 106 of 106	Held
P7	Open loop, 12 batches: at least 90 % of F rising edges of the stopper, output and inspection sensors have an M edge within 10 ms, and median chute-sensor rise difference $M - F > 0$	wrong if either fails	480 of 480; +170 ms	Held
P8	For each batch meeting P1, count the P8 settings (of 8) where F [PASS] $\wedge$ M [FAIL]; a batch "holds" if $8h \geq 5k$; more than half of the batches hold	wrong if not	3 of 3 hold ($h = 6, 6, 7$ of $k = 8$)	Held

No.	Falsification condition	Claim it would falsify	Result
F1	No correct-ladder pair of cell B (25) with $F \neq M0$	The model changes this cell's correct-ladder verdicts	Not triggered (4 of 25)
F2	Main-metric discordant pairs ≥ 2	Defect detection is nearly the same in both models	Not triggered (0)
F3	No correct-ladder pair of cell B with F [PASS] $\cdot$ M1 [FAIL]	Reversals remain after re-derivation	Not triggered (4 of 25)

How much each held prediction says differs. P2, P4 and F2 could plausibly have failed (the pre-registration flagged P2 as risky). P3 holds largely by construction, because M1 does not change the

chute capacity that a chute-full lock depends on. P8 is weaker than it looks: two of the eight settings made **all 12** batches fail under M0, including the 8 that agree at baseline (Section 5.4), so a setting counted as "holding" a reversal does not always single that batch out. Leaving those two settings out of k, h would be 4, 4 and 5 of 6 and all three batches would still hold ($8h \geq 5k$); counting them as not holding with k = 8, only 3570567229 would hold and P8 would be wrong. This recount is not pre-registered. P1 counted 3 rather than 4 because the fourth five-defect reversal locked on the chute confirm window, not on chute full; F1 and F3 count it.

The round was valid: 607 of 607 runs [VALID], no [PLANT_ERROR] or [INVALID], no retried run, every planned count met (main 162, second runs 106, gate 9, open loop 14, P8 156, model-assumption $60 = 10 \times (4 + 2)$, E4 100), no reference-match difference before a first lock.

5. Results

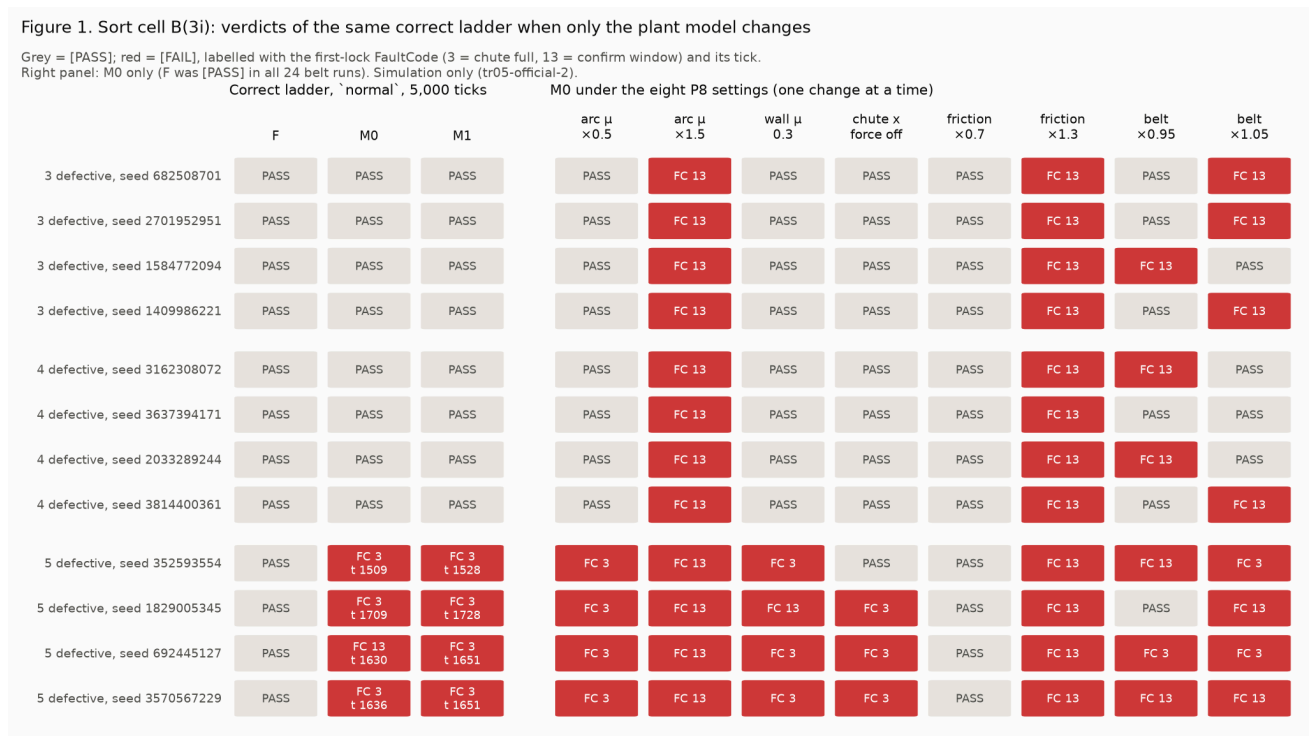


Figure 1. Sort cell B(3i), correct ladder, normal: verdict per batch under F, M0 and M1 (left), and the M0 verdict under the eight P8 settings (right). Red cells carry the first-lock FaultCode (3 = chute full, 13 = chute confirm window) and, on the left, its tick. Simulation only; tr05-official-2.

5.1 Correct ladder, sort cell B(3i)

Seed (defective parts)	F	M0	M1	Ledger-free M0 / M1	First differing event (M0); pre-mapped row
352593554 (5)	[PASS]	[FAIL] lock 3 at tick 1509	[FAIL] lock 3 at 1528	[FAIL] / [FAIL]	lock, chute full; r5 queue cap and brake ("different")
1829005345 (5)	[PASS]	[FAIL] lock 3 at 1709	[FAIL] lock 3 at 1728	[PASS] / [PASS]	lock, chute full; r5
692445127 (5)	[PASS]	[FAIL] lock 13 (chute window) at 1630	[FAIL] lock 3 at 1651	[FAIL] / [FAIL]	lock, chute confirm window; r3 pusher contact ("contact computation")
3570567229 (5)	[PASS]	[FAIL] lock 3 at 1636	[FAIL] lock 3 at 1651	[FAIL] / [FAIL]	lock, chute full; r5
the other 8 batches (3 or 4 defective)	[PASS]	[PASS]	[PASS]	[PASS]	—

- All four reversals are F [PASS] → M [FAIL], class (a), and deterministic (two identical runs each). No reversal in the opposite direction occurred.
- In the four M0 runs every part arrived (Arrivals 20), there was no physical event and no bridge judgment, all 5 defective parts were at route 1, and no part was on the floor or stacked. After the lock, 6, 2, 2 and 3 good parts remained at route 0 (9, 13, 13 and 12 at the output). At the chute-full locks the chute sensor was on.
- Under M1 the wider chute window removed the window lock of 692445127, but that batch then locked on chute full 21 ticks later; the other three locked 19, 19 and 15 ticks later than under M0 with the same code. Re-deriving the three constants moved the locks; it did not remove any. In positive control 2, run after the freeze on development seed 2617252722 (five defective parts) and not counted, M0 failed on the chute confirm window and M1 passed: there the re-derived window removed the reversal.
- curtain: 11 of 12 [PASS] and one [FAIL] (seed 1584772094) in each of F, M0 and M1 — not a reversal. In that batch the pusher never fired within the 1,000 ticks, so the operator never entered and the safety verdict, which requires a detected curtain event, could not pass (read from the raw tick records, which are not in the public files). recovery (one batch): [PASS] in all three.

5.2 Known defects

Ladder (pairs)	Expected-reason detection F / M0 / M1	Verdict [FAIL] F / M0 / M1
cell A late-reverse, swapped-sensors (4)	4 / 4 / —	4 / 4 / —
push-timing-off (12)	12 / 12 / 12	12 / 12 / 12
ignore-curtain (12)	5 / 5 / 5	12 / 12 / 12
no-sort-check (1, secondary only)	—	1 / 1 / 1
Main metric (28; M1 24)	21 / 21 / 17	
Secondary, 13 pairs failing by run length / other 16 (M1 12)		13 / 13 / 13; 16 / 16 / 12

The 7 ignore-curtain batches without the expected reason were the same 7 in F, M0 and M1. In 5 of them the pusher never fired within the 500 ticks, so the operator never entered; in the other 2 the first pusher command, and so the operator's entry, came at tick 496 (F), 497 (M0) or 498 (M1), leaving 2 to 4 ticks — less than the 0.265 s the bridge allows before it judges. These misses come from the scenario's trigger, not from the plant model (raw tick records; not in the public files). Under push-timing-off the first lock was FaultCode 13 in all 12 batches under F and M0; under M1 it was 13 in 7 and 5 (arrival sensor on too long, a constant that was not re-derived) in the 5 batches 1584772094, 3162308072, 3637394171, 2033289244 and 3814400361 — still detected by the expected reason.

5.3 Open loop (F requests replayed on M0, no ladder)

All 12 batches: 40 of 40 rising edges of the stopper, output and inspection sensors within 10 ms each (480 of 480), final routes 20 of 20, no chute-confirm age outside the original window. The 48 paired chute-sensor rises were later under M0 by 10 ms (19 pairs), 20 ms (1), 170 ms (9), 180 ms (2), 240 ms (3) and 390 ms (14): median +170 ms. Cell A (2 replays): route 1 of 1, no PART_DROP.

5.4 P8 sensitivity (one value changed at a time, M0 unless noted)

Setting	Correct normal, 12 batches: [PASS] / [FAIL]	Of the 4 reversed batches still [FAIL]
Arc friction × 0.5	8 / 4	4
Arc friction × 1.5	0 / 12 (all FaultCode 13)	4
Wall friction 0.3	8 / 4	4

Setting	Correct normal, 12 batches: [PASS] / [FAIL]	Of the 4 reversed batches still [FAIL]
Chute x force off	9 / 3	3
Friction × 0.7	12 / 0	0
Friction × 1.3	0 / 12 (all FaultCode 13)	4
Belt speed × 0.95 (F: 12 / 0)	6 / 6	3
Belt speed × 1.05 (F: 12 / 0)	4 / 8	4

Two settings (arc friction × 1.5, friction × 1.3) turned every batch into [FAIL], and the belt settings failed 3 and 4 of the 8 batches that agree at baseline; friction × 0.7 removed all four reversals. The verdict is thus sensitive to assumed values in both directions, and P8's "holding" partly counts settings that fail everywhere. In cell A the 6 pairs gave the baseline verdicts (correct [PASS], negatives [FAIL]) under every cell-A setting run (friction × 0.7 and × 1.3 in M0, belt × 0.95 and × 1.05 in F and M0).

5.5 Model-assumption settings (reported only, E8)

Batches: the 4 reversed and the first non-reversed batch of the 3- and 4-defect bins. Eight M0 settings each (rotated sensor footprint, step 0.0005 s, 20 arc facets, deck static-friction weight 0, torsional friction 0 and 0.02 m, lateral start ±5 mm) and F at ±5 mm.

Batch	M0 [FAIL] in (of 8)	Settings where M0 passed	F at ±5 mm
352593554	7	step 0.0005 s	[PASS]
1829005345	7	20 arc facets	[PASS]
692445127	7	step 0.0005 s	[PASS]
3570567229	8	—	[PASS]
682508701 (3, not reversed)	1 (torsional 0.02 m)	7 others	[PASS]
3162308072 (4, not reversed)	2 (deck static weight 0; −5 mm)	6 others	[PASS]

Under the pre-registered rule, 352593554 and 692445127 are marked "possibly numerical", because halving the step removed their reversal.

5.6 Other exploratory results (no hypotheses)

- **E6 Cell A correct ladder** (seen before registration): [PASS] under F and M0 in normal and cycle-stop.
- **E4 TR-2026-03 cell A interlock-removal variants** (25 × 2 scenarios): F and M0 gave the same verdict in all 50 pairs (12 variants [FAIL] in both scenarios, 13 [PASS] in both).
- **E2 Event-code differences.** Beyond the reversals, the only first-lock difference among negative ladders is the M1 push-timing-off FaultCode 5 above. The other parts of E2 (full event-code sets), E3 (rotation report) and the per-tag sensor-edge statistics of table 3 were not computed by the fixed scripts and are not reported; the raw records are kept.
- **Reference match.** In the 135 correct-ladder runs, no difference before the first lock; the documented one-scan exception appeared in 45 runs, always at tick 4461. After the first lock, differences in 4 runs (both runs of seed 1829005345 under M0 and M1): 13,130 scans in total, all on B.Lamp.red and B.Lamp.yellow.
- **E5 Run cost** (wall clock per tick, sequential runs, from the raw run summaries): correct normal, median F 1.81 ms (1.44–2.11), M0 9.55 ms (5.47–10.71), M1 9.42 ms (5.81–10.41). The whole round took 2 h 55 min.

6. Run ledger and every change after the prediction freeze

Time (KST, 2026-10-08)	Step	Result
13:33:43	Prediction freeze: v0.6 frozen sections committed (3a12187)	before any calibration, seed table or official run
after freeze	Positive control 1; seed table (official seeds loaded only, no tick run); calibration (6 F + 6 M0 replay runs); M1 ladders	passed; constants in Section 3.4
after freeze	First positive control 2 — invalid, kept. Development seed 2617252722 under M0: the ladder locked on the chute window at tick 1697 (fifth defective box never reached the chute sensor); the original reference logic compared the same wait in floating point ($2.3500000000000014 > 2.35$) and locked one scan earlier, at 1696, which invalidated the control under the reference-match rule as written	folder moved to _invalid/, not deleted; no tables made
13:55:28	Amendment 1 committed (1ac8c13): for F and M0 the reference logic also uses the half-scan inward margin already registered for M1 (chute window 2.0 / 0.3525 s, output window 2.335 / 0.3575 s, chute full 0.445 s); constants, ladders, plant, verdict rules, predictions and analysis unchanged	before any official run
before 14:17	Runner changes outside the frozen sections: development-seed path of the round runner, wider fixed-input file, amendment-1 implementation; the source list carries the hash of a draft fixed-input file (circular dependency, declared in v1.0); positive control 1 re-run and positive control 2 re-run in a new folder	positive control 2 valid: F 3 of 3 [PASS], M0 2 of 3 (2617252722 [FAIL], lock 13 at 1697), M1 3 of 3
14:17:32	v1.0 committed (1ee07b3); frozen sections byte-identical to the freeze	
14:19:16–14:23:13	tr05-official-1 — invalid. A line added to the round runner for positive control 2 turned the else of the batch selection into the else of the new if, so in official mode the 12 batches were replaced by an empty development list and no sort-cell run happened ; the end-of-round count check computed the expected counts from the same empty list and passed. 159 cell A runs ran, all [VALID]; void under the frozen run-count rule. Outcomes seen there (official inputs): cell A correct ladder [PASS] and both negative ladders [FAIL] in both scenarios under F and M0; repeats identical; open loop route 1 of 1, no drop; cell A P8 M0 [PASS] 8 / [FAIL] 16, belt F [PASS] 4 / [FAIL] 8; E4 F and M0 [PASS] 26 / [FAIL] 24	folder moved to _invalid/; no prediction computed
14:29:28	Correction 1 committed (cd0d54a): runner batch selection restored; official mode now refuses to start unless the plan equals the frozen counts; round.py 90cfe55c... → 0649b9fd..., source list 88d0920e... → 49b7f47d..., fixed-input file 532167ac... → e7566884...	second round under the frozen retry rule
14:30:33–17:25:56	tr05-official-2 — valid. 607 of 607 [VALID], every count met. All 159 cell A runs gave the same results as in round 1 (157 with identical verdict and trace hash; the 2 open-loop replays identical except wall-clock time)	the only round counted
after the results; committed 18:04:16 (227 8552)	Correction 2 , recorded after the official results (document SHA-256 a6e747c770b7a5e26026c43cf3f70c8e15ca41b78deb1a68ff03f3512480f7b; its PREREGISTRATIONS.md row says it was recorded after the results). The independent recompute stopped with UnboundLocalError in its P8 path on its first run (development rounds had no P8 batch, so the line never ran). One line (line 137) was changed: recompute.py a5e694044937926ddf87bd7fb633b738ca50b0ec9d7d3a4d5c73e40de70a856b → 0e90413be91901005e20ec2fc7ca189bfb0824454147321e1b7b7f9196282dd6. The runner, the table script and the public conversion did not change. The fixed-input file still lists the old hash, so recompute.py in the bundle differs from fixed.json; the old file is released beside it	after the fix, recompute on the raw-round tables and on the public tables: 0 differences each

There was no other official round.

7. Threats to validity and what this report does not claim

- This is a simulation. Neither model was compared with a physical cell, PLC, conveyor, sorter or box. **The report does not claim that the MuJoCo model is more correct**, only that changing this bundle changed these verdicts.
- [PASS], [FAIL] and "detection" are values of the verdict rules inside the simulator. They do not say that a ladder is correct or safe on equipment, and no plant-connected test in either model replaces commissioning. Detection counts are for the five negative ladders chosen here and describe no product.
- One machine, one build, one MuJoCo library. Determinism and the comparison are claimed only there.
- **Cell B(3i) is not the cell B of TR-2026-03 and TR-2026-04** (Section 3.3). Nothing here says how sensitive the 3k cell's verdicts are to the plant model.
- **Cell A correct-ladder results are exploratory**: the identical input had been run under M0 before registration, and its official outcome was also seen in the invalid first round before the second round.
- All four reversals are in the five-defect bin, and the bins are a design choice (12 batches drawn by a seed rule and binned by defect count), not a sample of a defect rate of 0.2. No confidence interval is given.
- The reversals are sensitive to assumed physical values (Section 5.4): $\pm 30\%$ changes of all sliding friction either remove all of them or make every batch fail. Two of the four disappear at half the time step. The MuJoCo-side values (guide and wall geometry, zero wall friction, static-friction weight, the two 2 mm tolerances, facet count) were assumed, not measured; the 2 mm tolerances were not varied.
- Sensors use an axis-aligned box at the part centre; rotated parts are not seen as rotated except in one setting. The chute x force acts at the centre of mass and lacks the yaw damping of real anisotropic friction, which tends to increase yaw on the chute. The eight P8 settings do not include the most direct causes of yaw (sensor footprint, plate friction, deck kinetic friction); the footprint is varied only among the model-assumption settings (Section 5.5) and plate friction not at all.
- recovery and no-sort-check ran on one five-defect batch only, so under M they can overlap with a chute-full lock.
- CENTER-ENTRY cannot occur in M, and no positive case of CENTER-SNAP was tested in either model.
- M1 is this study's re-derivation rule, not a reproduction of an original fitting procedure (none is documented); it does not change chute capacity, so P3 was expected to hold by construction.
- Actuators are kinematic in both models (no cylinder force or stiffness limit, no plate dynamics), so stroke-time faults cannot differ between models.
- MuJoCo's solver reaching its iteration limit without convergence raises no warning, so it would not appear as [PLANT_ERROR].
- The 7 missed ignore-curtain detections and the one failed correct curtain run come from the scenario trigger (no pusher command in time), identically in every condition.
- E2 (in full), E3 and the per-tag edge statistics were planned but not computed by the fixed scripts.
- The independent recompute recounts the same per-run files with separately written code; it does not measure anything by another method. It was itself changed after the results (correction 2).
- The original methodology review of 2026-09-29 that the pre-registration summarises could not be found on the machine used; the pre-registration says so.

8. Released files, and what a reader can and cannot check with them

File	Content	Licence
TR-2026-05.md, TR-2026-05.pdf	This report	CC BY 4.0
TR-2026-05/MANIFEST.md	SHA-256 and licence of every bundle file	—
TR-2026-05/preregistration/	Prediction freeze v0.6, amendment 1, fixed version v1.0, correction 1 and correction 2 (their SHA-256 are in PREREGISTRATIONS.md; correction 2 was committed after the results), and the rule that cuts the frozen sections (Korean)	CC BY 4.0
TR-2026-05/controls/	Public version of the positive-control and calibration summaries (positive control 1 before and after amendment 1, the calibration runs, the invalid first positive control 2, positive control 2 after amendment 1 with its tables, recompute and public version, the patent-check results, logs), with local paths replaced by tokens listed per file in MANIFEST.md; and the two positive-control references (the per-seed sort results of the 3i fast model and the macOS 3i reference verification summary), byte-identical	CC BY 4.0 (calib_runs.py and chain_a1.sh: MIT)
TR-2026-05/fixed/inputs/	Seed table, fixed-input file fixed.json, source list, M1 constant table, M1 reference parameters, regeneration report, calibration record (calibration_public.json: local paths replaced by calib/; the original's SHA-256 is in MANIFEST.md)	CC BY 4.0
TR-2026-05/fixed/bridge/	Runner and analysis scripts listed as public in the pre-registration (run.py, tr05.py, transport.py, round.py, seed_table.py, calibrate.py, regenerate_m1.py, tables.py, recompute.py, public.py, patent_check.py, common.py, refcheck.py, classify.py, replay.py, probe_engine.mjs) and recompute_as_fixed_a5e69404.py (the recompute as fixed, before correction 2)	MIT
TR-2026-05/ladders/	Public versions of the cell A correct ladder and of the cell B(3i) correct ladders (original and M1) = the run version with line 4 (the marker field, as described in TR-2026-03, Section 8) removed; re-inserting that line gives the run SHA-256 in ladders-sha256.json	CC BY 4.0
TR-2026-05/results/	Public version of the round: results.jsonl (607 rows), tables.json, runs/*.json (607 per-run digests), counts.json, plan.json, traces-sha256.txt, the string-scan result, the two recompute outputs and the pre-registered patent keyword check (patent_check_official.json)	CC BY 4.0
TR-2026-05/experiments/S2/criteria-3i.md	Expected reasons of the 3i negative ladders, byte-identical	CC BY 4.0
TR-2026-05/scripts/figure1.py, TR-2026-05/figures/figure1.png	Figure 1 from results/	MIT, CC BY 4.0

A reader can (from the repository root after `mkdir out`, with `PYTHONDONTWRITEBYTECODE=1` so that Python writes no `__pycache__` into the bundle): hash the pre-registration files and compare them with `PREREGISTRATIONS.md` (the frozen-section hash with the cutting rule); rebuild every table and verdict from the public files (`python3 TR-2026-05/fixed/bridge/tables.py TR-2026-05/results --out out/tables.json` gives a file byte-identical to `results/tables.json`); run the independent recompute (`python3 TR-2026-05/fixed/bridge/recompute.py TR-2026-05/results --tables TR-2026-05/results/tables.json --out out/rc.json`: 0 differences, 593 runs); see the defect of the recompute as fixed (`recompute_as_fixed_a5e69404.py` stops with `UnboundLocalError`); check the positive-control and calibration results of Sections 3.4, 3.7 and 6 against `controls/` (for example the invalid first positive control 2 in `controls/tr05-calib-1/p c2/`); and redraw Figure 1 (`python3 TR-2026-05/scripts/figure1.py TR-2026-05/results out/figure1.png`, needs `matplotlib`; visually identical, not byte-identical). A reader cannot rerun the simulation, recompute the calibration or the physics hashes, or confirm that the files came from the stated

commit: the application, its transport code and scene files, the cell, I/O-map and device files, the reference logic, the ladder generator, the modified ladder engine, the cell B(3i) negative ladders (SHA-256 only) and the tick traces (about 45 GB; their SHA-256 are listed) are not released. The cell A negative ladders are the files already released with TR-2026-02. The pre-registration, the scripts, the fixed inputs and the control metadata name the company's private repositories, branches and internal paths (as in the TR-2026-02 and TR-2026-03 bundles); they are not accessible. The patent-check passage of the pre-registration (section 13-1), the patent-check script and its result files in controls/ name the application KP26038 and keywords of its claim; these were already published in the TR-2026-04 pre-registration v0.4, Section 13, and this report does not disclose them for the first time. The raw-round tables.json differs from the public one only in its calibration table, which carries local paths and is replaced by fixed/inputs/calibration_public.json.

9. Checks before release

Gate	Status
Pre-registration hashes: v0.6 (whole file and frozen sections), amendment 1, v1.0, correction 1 and correction 2 equal the committed values	[PASS] (re-hashed in the bundle)
Invalidation criteria of the counted round	[PASS] round valid
Independent recompute on the raw-round and public tables	[PASS] 0 differences each, after correction 2
Correction 2 recorded publicly	[PASS] PREREGISTRATIONS.md row committed after the results (2278552, 18:04:16 KST); document byte-identical in the bundle
Isolated reproduction from MANIFEST.md (tables byte-identical, recompute 0 differences, every listed SHA-256)	[PASS] in a copy of the bundle
Numbers in the text against the result files	[PASS] re-extracted by script after the review fixes
Patent keyword check on the official bundle (planned in the pre-registration, §13-1): patent_check.py with the TR-2026-08 keyword tool (positive control 16 hits) over the whole bundle, this report and the README text	[PASS] no word-level hit in the report or the result files (the report's only raw matches are an English keyword of the tool inside the word "paragraph" in this row; the result files are clean apart from the check's own record results/patent_check_official.json, which repeats the keywords it found); the pre-registered command (22 targets: public scripts, pre-registration v1.0, tables.json, ladders) found only the composition sentence of pre-registration §13-1 (5 keywords, twice) that lists the five patent features as absent; over the whole bundle only that sentence (v0.6 and v1.0), its copies in patent_check.py and the patent-check records under controls/, and the self-test strings of patent_check.py; two false positives in the README text (the same keyword inside the word "paragraph")
Public-version string scan of the full bundle	[PASS] with disclosure: the pre-registered criterion is 0 hits (pre-registration §12); the only hits are 8 strings inside the hash-fixed pre-registration files (home-relative tool and run paths; two mentions of the confidentiality scan term in the sentence that defines the scan), released unchanged by decision because editing them would break the registered hashes and listed by file and line in MANIFEST.md, and 1 false positive in the PNG bytes
Pre-release review by AI agents, given the text as an outside draft (recount of every number from the public files, pre-registration consistency, claim scope, anonymisation, reproducibility, licence)	[PASS] after fixes: first review 6 must-fix applied, except that the non-filing sentence was kept to the scope approved by the author (MuJoCo part-transport rules and model) and not widened to the M1 re-derivation rule, which was left to the author; its section references were narrowed; 10 of 11 recommendations applied (the remaining one is a repository note changed at release); numbers 107 of 108 matched, the one mismatch (Section 3.5) corrected; final review 2 must-fix (wording of this table), applied, with 52 values recounted from the public files and 0 mismatches

Gate	Status
Company, person and legal statements (author name, affiliation and e-mail, company location and legal name, the two patent numbers as in TR-2026-07, licence and disclosure sentences, third-party names), checked against the published reports TR-2026-06 and TR-2026-07 and the company's rules file; the company's source-of-truth folder was not available on the machine used	[PASS] after applying the fact-checker's wording verbatim
Forbidden-term sweep	[PASS] 0 findings (report, README text, MANIFEST)
Numbers in the PDF against the Markdown	[PASS] 200 distinct numbers, none missing
Author's release confirmation	given 2026-10-08 before the official round (publish whichever way the result comes out)

10. Use of generative AI

LLM coding agents (Claude, Anthropic; earlier stages also Codex, OpenAI) wrote and changed the MuJoCo transport code during development, wrote the runner, calibration, table, recompute and public-conversion scripts, drafted the pre-registration, its amendment and corrections and this report, and ran the calibration and the official rounds. AI agents that did not read each other's output reviewed the pre-registration drafts (design, arithmetic, public-release risk, domain) and the application code, and further AI agents reviewed this report before release (a pre-release review and a fact check). No person reviewed the code line by line. The author instructed on 2026-10-08 that the study proceed (「tr-05는 진행함」) and approved, before the official round, the publication of the report, the public result files and the runner scope whatever the result, and the disclosure of the transport rules without a patent filing (「추천대로」). The author is responsible for the content.

Licence

Code (every .py, .mjs and .sh file in the bundle) is released under the MIT licence, Copyright (c) 2026 WACE Inc. (주식회사 웨이스); the report, tables, figure, results, the pre-registration documents and every other file of the bundle that is not code are released under CC BY 4.0. The licence of each file is declared in MANIFEST.md, and the licence texts are in LICENSES/ at the repository root. This release does not grant any licence under patents of WACE Inc., including Korean patents No. 10-2904772 and No. 10-3003470, or under any pending or future patent application of WACE Inc. No patent will be filed on the MuJoCo part-transport rules and model described in Sections 3.1 and 3.2 and Appendices B and C (transport rules v1 and v4, the physics values lists and the scene construction); this report serves as their disclosure. Other names in this report (MuJoCo, NVIDIA GB10, Ubuntu, Linux, macOS, Node.js, Clang, Python, matplotlib, Claude, Codex) belong to their respective owners; this report is not affiliated with them. MuJoCo (Apache-2.0) is not included in the bundle; its version and library SHA-256 are given.

Appendix A. Where each number comes from

Number	Source
P1 to P8, F1 to F3 values and verdicts	results/tables.json → predictions (written by tables.py, reproduced by recompute.py)

Number	Source
Section 5.1 table, first differing events, mapped rows	results/tables.json → tables.table1, tables.table6; parts, arrivals and lock flags from results/runs/main.<cond>.cell-b__correct__normal__s<seed>__r1.json
Section 5.2	results/tables.json → tables.table2, outcomes (first-lock FaultCode fc)
curtain and ignore-curtain trigger ticks	raw tick records (trace.jsonl, not released; SHA-256 in results/traces-sha256.txt)
Section 5.3	results/tables.json → tables.table3
Sections 5.4 and 5.5	results/results.jsonl (groups p8, model), first-lock codes from results/runs/p8.*, model.*; P8 h and k from tables.table4
Reference match	results/tables.json → tables.table7
Calibration (Section 3.4)	fixed/inputs/calibration_public.json, fixed/inputs/m1-constants.json
E4	results/results.jsonl (group e4)
E5	raw run summaries (timing.ms_per_tick), not released
Run counts, times	results/counts.json, round log (start and end lines; not released)
Positive controls, calibration runs, invalid first positive control 2	controls/tr05-calib-1/ (gate/, gate-a1/, calib/, pc2/, pc2-a1/), fixed/inputs/calibration_public.json
Round 1 comparison	round 1 results.jsonl and open-loop summaries (not released; listed in correction 1)

The condition names in the files are F, M0 and M1; groups are main, repeat, gate, openloop, p8, model and e4.

Appendix B. Physics value lists

Defaults in commit ac78d02. Source: assumed (no measurement or device data), device data (copied from the device definition), derived (from a fast-core rule or a measurement), switch (sensitivity only; no effect at its default). Part dimensions, masses, friction and restitution, belt, chute and arc friction and gravity come from the cell and device definitions and enter values_sha256 through the scene.

Cell A, transport rule v1 (15 values)

Name	Default	Meaning	Source
fixed_step_s	0.001 s	MuJoCo fixed step	assumed
deck_thickness_m	0.02 m	deck box thickness	assumed
guide_height_m	0.03 m	side guide height above the deck	assumed
guide_thickness_m	0.01 m	side guide thickness	assumed
guide_clearance_m	0 m	gap between deck edge and guide	assumed
floor_thickness_m	0.05 m	floor box thickness	assumed
floor_size_m	20 m	floor box side	assumed
friction_static_weight	1	part friction = $w \cdot \text{static} + (1 - w) \cdot \text{kinetic}$	assumed
torsional_friction_m	0.005 m	torsional friction (condim 6)	assumed
rolling_friction_m	0.0001 m	rolling friction (condim 6)	assumed
deck_restitution	0	deck contact restitution	assumed
guide_restitution	0	guide contact restitution	assumed
floor_restitution	0	floor contact restitution	assumed

Name	Default	Meaning	Source
tilt_flag_rad	0.0349 rad	report flag above 2° tilt	assumed (report)
friction_scale	1	scale of all part contact friction	switch

Cell B(3i), transport rule v4 (32 values)

Name	Default	Meaning	Source
fixed_step_s	0.001 s	MuJoCo fixed step	assumed
deck_thickness_m	0.02 m	deck box thickness	assumed
guide_height_m	0.03 m	side guide height	assumed
guide_thickness_m	0.01 m	side guide thickness	assumed
guide_clearance_m	0 m	gap between deck edge and guide	assumed
floor_thickness_m	0.05 m	floor box thickness	assumed
floor_size_m	40 m	floor box side	assumed
deck_friction_static_weight	1	deck-part friction = $w \cdot \min(\text{belt, part static}) + (1 - w) \cdot \min(\text{belt, part kinetic})$	assumed
chute_friction_static_weight	0	same blend for the chute slope (0 = kinetic)	assumed
part_friction_static_weight	1	part-part friction blend	assumed
wall_friction	0	friction of guides, centering walls, chute walls, end wall, output cushion	derived (fast core clamps without friction)
floor_friction	0.5	floor-part friction	assumed
torsional_friction_m	0.005 m	torsional friction of all contacts	assumed
rolling_friction_m	0.0001 m	rolling friction of all contacts	assumed
static_restitution	0	restitution of static surfaces and the plate	assumed
transition_facets	10	chord facets of the arc	assumed (discretisation)
transition_thickness_m	0.02 m	arc facet and chute slope thickness	assumed
chute_wall_thickness_m	0.01 m	chute side and end wall thickness	assumed
centering_wall_bottom_m	0.015 m	centering wall bottom above the deck	device data
centering_wall_height_m	0.02 m	centering wall height	device data
centering_wall_thickness_m	0.008 m	centering wall thickness	device data
output_stop_bottom_m	0.0025 m	output cushion bottom	device data
output_stop_height_m	0.1375 m	output cushion height	device data
output_stop_thickness_m	0.01 m	output cushion thickness	device data
plate_thickness_m	0.012 m	pusher plate thickness	device data
center_contact_tolerance_m	0.002 m	CENTER-SNAP penetration allowance (measured maximum 1.66 mm)	derived (measured)

Name	Default	Meaning	Source
output_contact_tolerance_m	0.002 m	output arrival: $x \geq$ fast output cap – this value	assumed
tilt_flag_rad	0.0349 rad	report flag above 2° tilt	assumed (report)
friction_scale	1	scale of all sliding friction and of the chute x force target	switch
transition_mu_scale	1	scale of arc friction and of the arc share removed by the chute x force	switch
chute_x_force	1	1: chute x force on (rule v3 behaviour); 0: isotropic friction only	switch
sensor_footprint_rotated	0	1: sensors see the bounding box of the rotated part in the sensor frame	switch

Appendix C. Differences between the two models

Row	Feature	F	M	Same?
r1	Belt transport	x += belt displacement	surface velocity + contact friction	replaced by contact computation
r2	Part rotation	no degree of freedom	free; not fed to default sensors	different
r3	Pusher pushing	kinematic transfer step	plate-box contact	replaced by contact computation
r4	Chute x friction 0.85	x deceleration rule	isotropic friction + chute x force at the centre of mass	approximation
r5	Chute queue cap, spacing, end brake (last 0.9 m, 5 m/s ²)	rules	end wall and box-box contact only	different
r6	Chute jam	chute parts stop	CHUTE - HOLD keeps their pose	moved to give the same effect
r7	Chute capture (route 1)	centre past the lip	same criterion, tick end	same
r8	Output arrival (route 2)	aligned to the cap	contact stop within 2 mm of the cap; not a sensor input; the queue can buckle	tolerance added
r9	CENTER-ENTRY	formula	same formula, cannot occur (guides)	out of scope
r10	CENTER-SNAP	front-corner formula	deepest footprint corner – 2 mm	approximation
r11	FLY-SIDE, PUSH-RETURN	event only, plate passes through	same event, plate hits the box	same event, different outcome
r12	FLOW-QUEUE	formula	same formula, also with real contact and rotation	same event condition
r13	Collection (route 1 → 4)	collector kinematics	same, held until clear, then inactive	same
r14	PUSH_MISSED, sensor logic, HMI counts	core	same code	same
r15	PART_DROP (cell A)	deck-end rule	same rule + side drop	same (+ secondary)
r16	Fall to the floor (cell B)	cannot occur	can occur, reported only; part stays route 0	different
r17	Stacking	cannot occur	can occur, reported only; sensors may miss a stacked box	different

Row	Feature	F	M	Same?
r18	Cell A guide and floor friction	—	part coefficient via max combination	assumption
r19	Floor height	0	0	same
r20	friction_scale	not read (F bytes unchanged)	scales friction	sensitivity only
r21	Lateral start position	cell B read; cell A 0	cell B same; cell A refused if not 0	cell B same, cell A no model